

After Surprise: Exploratory Authorship and the Temporal Structure of Generative Making

Watson Hartsoe

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Abstract

Debates about human authorship in generative systems often treat control as a question of prior specification: did a person determine the expressive result before the machine produced it? That framing fits some kinds of making but poorly describes exploratory practices in which intentions and even the relevant dimensions of choice emerge only after an intermediate result is perceived. Current U.S. Copyright Office analysis distinguishes human conception from machine execution and treats repeated revision under present systems as insufficient control. A contrary scholarly position argues that authorship can arise through iterative exploratory engagement. This article develops a narrower bridge between those positions. It argues that surprise itself is neither authorship nor its negation. The evidentiary event is prospective stabilization: an initially unanticipated feature becomes a newly formed human commitment and then causally governs subsequent interventions. This criterion distinguishes exploratory making from mere search more sharply than iteration count, effort, or unpredictability. Video-game cases show why causal influence over a path is insufficient by itself; design theory explains how new controllable distinctions emerge in use; empirical work on human adaptation to generative systems shows why final instructions cannot transparently reveal original intention. The result is a temporal account of agency that asks not whether the maker predicted the first result, but whether a discovered difference could be made to matter to what happened next.

Keywords: authorship; generative systems; exploratory design; human–computer interaction; control; copyright; reflective practice

1 Introduction

The most familiar question in disputes over computer-generated expression is whether the human user controlled the result. The question sounds precise but hides a temporal assumption. It imagines a finished conception upstream and asks whether the downstream artifact faithfully realizes it. That sequence is visible in contemporary copyright analysis, which distinguishes ideas or mental conceptions from the fixed expressive elements produced by a generative system. The U.S. Copyright Office currently concludes that generally available systems do not let textual instructions alone determine those expressive elements with sufficient control, and it treats repeated revision as additional sampling rather than as a different kind of execution.[1]

That framing fits cases in which a person knows the relevant expressive variables before execution. It fits less comfortably when making is exploratory. A painter can discover a relation only after a mark is on the canvas. A photographer can encounter an unforeseen event and then change position, exposure, or timing in response. A designer can see a quality in an intermediate configuration that was not available as an explicit variable at the beginning. In these cases, the history of authorship cannot be reconstructed solely by asking what was specified in advance.

The dispute over generative systems therefore needs a temporal distinction more discriminating than predictable versus unpredictable, deterministic versus stochastic, or human versus machine. Surprise itself is neither proof of authorship nor proof against it. The stronger question is what happens after surprise. When an unanticipated feature appears, does the person merely accept it, or does that feature become a prospective commitment that structures later interventions? I call the latter event prospective stabilization.

Prospective stabilization offers a narrow bridge between two positions that otherwise seem opposed. The Copyright Office is right that effort, repetition, and causal influence are not enough. Scholars who defend the possibility of iterative authorship are right that a complete conception need not precede every authored work. The missing discriminator is whether a newly perceived difference becomes a condition on what the maker does next. This is not proposed as a replacement for copyright doctrine. It is an evidentiary account of creative agency that can be tested against production histories.

2 The ex ante control model

The Copyright Office’s 2025 report states that, given current generally available technology, textual instructions alone do not provide sufficient control over generated expressive elements. The Office emphasizes the gap between a user’s conception and the system’s conversion of that conception into fixed expression. It also rejects the idea that repeated revision changes the analysis when revision simply causes the system to produce more outputs from which a user selects. The legal concern is not unpredictability by itself; the report says the relevant issue is the degree of human control.[1]

This analysis has a recognizable legal ancestry. In *Andrien v. Southern Ocean County Chamber of Commerce*, the Third Circuit distinguished authorship-preserving embodiment from an intermediary contribution that requires intellectual modification or highly technical enhancement. The delegating party may remain the author when the embodiment is rote or mechanical.[8] The contemporary question is therefore not merely whether a machine lies between conception and artifact. It is what expressive work remains unresolved when the intermediary acts.

The Office’s approach is expressly technology contingent. It leaves open a future system in which human control becomes sufficiently fine grained that the system’s contribution approaches rote or mechanical execution.[1] That qualification matters because it prevents the current rule from becoming an ontology of language-based creation. The rule describes a claimed relation between present interfaces and present expressive control, not a permanent claim that natural-language direction can never participate in authorship.

3 Exploratory making and the production of intentions

A contrary submission to the Copyright Office by Pamela Samuelson, Christopher Sprigman, and Matthew Sag accepts that simple textual direction will often be insufficient but rejects a categorical boundary. More importantly, the authors defend a second route to authorship that does not depend on a fully formed initial conception. They argue that iterative engagement with a medium can involve contemplating alternatives, embracing some, and rejecting others as a work develops.[2] The significance of this claim is easy to miss if it is reduced to the number of iterations. Its actual target is the temporal structure of intention.

Donald Schon provides a much older and more exact account of that structure. In his analysis of design as reflective conversation, objects and relations in a design situation are not simply given as complete inputs at the beginning. Designers see, move, and see again. They discover unintended consequences in intermediate products, and their intentions evolve through the process. Schon explicitly warns against a retrospective error in which distinctions that emerged only at the end of a process are read back into its beginning.[3]

This observation changes the control question. Suppose an intermediate image contains a spatial relation that the maker had not imagined. At the moment it appears, the relation was not selected from a previously enumerated menu. Yet the maker can recognize it, decide that it matters, and thereafter use every available operation to preserve or transform it. A later account that asks whether the relation was specified at the outset will classify the event as accidental. A later account that looks only at the final artifact may classify it as intentional. Both miss the actual sequence. The feature began as a surprise and became a commitment.

The possibility space of creative action is therefore partly endogenous to making. A feature can become a variable only after an earlier operation makes it perceptible. This is why an inventory of initial instructions is not enough to reconstruct agency. One must also ask when each consequential distinction became available to perception and when it first began to govern subsequent action.

4 Why selection and causal influence are not enough

Exploratory accounts face an immediate problem. Consumers also discover preferences while searching. A person who browses hundreds of images can learn what they like, reject alternatives, and stop when something attractive appears. If preference formation alone were sufficient, search would collapse into authorship. The distinction therefore requires evidence stronger than liking, adoption, or accumulated effort.

Early video-game cases provide unusually useful negative tests. In *Midway Manufacturing Co. v. Artic International*, the Seventh Circuit acknowledged that player actions alter which images appear but rejected the claim that each realized sequence is authored by the player. The player cannot create any sequence desired; the game constrains play to sequences its system permits.[6] The point is not that players lack skill or causal influence. They may have both. The point is that causal trajectory control can remain navigation within an expressive space structured upstream.

Stern Electronics, Inc. v. Kaufman sharpens the same problem from another direction. Each play of *Scramble* varies with player action, but substantial visual and aural elements recur across performances. The court located protectable originality in those recurring elements and treated

the underlying program as a distinct work.[7] Variability therefore does not force a choice between complete predetermination and complete user authorship. A system can contain authored invariants while allowing another actor to steer realized paths.

These cases block a simple equation between counterfactual control and authorship. If moving a control changes what happens next, that fact establishes influence, not necessarily authorship. A better analysis separates at least three roles: navigating a possibility space, constraining the space in a way that introduces stable expressive relations, and modifying the machinery that generates the space. The boundaries among these roles are technically different in learned generative systems than in 1980s arcade hardware, but the conceptual warning survives: making the path go otherwise is not enough.

5 Prospective stabilization after surprise

The gap between prior specification and terminal selection can now be stated more precisely. Consider a feature u that first appears unexpectedly at time t_1 . At time t_2 the maker treats u as consequential. At later times the maker changes instructions, spatial conditions, parameters, masks, source material, or other controls so that u is preserved, deliberately transformed, or protected from changes elsewhere. Competing branches in which u disappears are actively corrected rather than merely ignored. The maker did not author the first appearance of u in the ordinary sense of prior determination. But the later persistence of u is no longer explained by the initial accident alone.

This sequence is prospective stabilization: discover, commit, intervene, stabilize. The key direction is prospective. A post hoc narrative that says “I meant that” after the process ends is weak evidence. A subsequent action that sacrifices other possibilities in order to preserve the discovered relation is stronger because it can be contradicted by the production history. If the claimed commitment was real, later choices should bear its cost.

The criterion also explains why iteration count is a poor proxy. Six hundred revisions can contain little prospective stabilization if each state is merely discarded until an acceptable result arrives. Two revisions can contain substantial agency if the first reveals a new relation and the second deliberately reorganizes the process around it. The difference is not length but causal thickness: whether a perceived distinction becomes a constraint on later reachable states.

This account does not require a maker to verbalize every intention. Behavioral evidence can be enough. After a feature first appears, does the person repeatedly preserve it while changing other dimensions? Do they restore it after a later operation damages it? Can they produce a neighboring state that changes one property while holding the newly valued relation stable? These are stronger receipts than retrospective description because they expose the claimed commitment to failure.

6 Effective control and the problem of the interface

Control itself should also be separated from the number of visible options. Work on empowerment in agent-environment systems defines effective influence in terms of the capacity of actions to produce distinguishable future states that the agent can sense.[5] The measure is not a theory of authorship, but it supplies a useful technical lower bound: a control surface is weak when apparently different

interventions collapse into indistinguishable consequences.

For creative systems, the relevant state representation cannot be raw pixels. It must preserve the distinctions that matter to the task: pose, lighting, composition, identity, contour, timing, or whatever the practice treats as consequential. Effective control is then observer relative. A one-word operation that reliably preserves nineteen properties while changing the twentieth can carry more practical leverage than a long textual description whose consequences remain globally entangled.

This helps clarify why interface design changes the authorship question without changing the user’s imagination. A system may expose only coarse global conditioning, or it may expose masks, retained state, pose constraints, regional edits, reusable references, and other channels that let a person make local differences survive. The same intention can therefore have different causal reach in different interfaces. Current doctrine already acknowledges the technology contingency; an evidentiary model should do the same.

7 Co-adaptation and the instability of original intention

Iterative interaction introduces another difficulty: the maker is not the only thing changing. Don-Yehiya, Choshen, and Abend find that users updating natural-language instructions in response to Midjourney outputs converge partly because they remember or discover missing details and partly because they adapt to language patterns preferred by the model.[4] The final instruction therefore cannot be treated as a transparent record of a pristine intention.

This is not necessarily a defect. Media shape practice. Painters learn paint, photographers learn lenses, and musicians learn instruments. Yet co-adaptation means that an evidentiary account must avoid two symmetrical mistakes. It should not assume that every new commitment was present in the person before interaction. It should also not assume that every newly learned distinction belongs to the person simply because it appears in later language. Some changes express discovery; some express accommodation to system defaults; often both processes occur together.

Prospective stabilization helps only partially. It can show that a distinction later governed human action, but it does not by itself establish where the distinction originated or whether it reflects a model-specific convention. That remaining attribution problem is substantive. A complete account would compare independent representations of intention, cross-system transfer, and the stability of the claimed distinction when the underlying model changes.

8 An evidentiary protocol

The argument can be turned into a testable protocol. First, record the production sequence rather than only the final artifact. Second, timestamp the first appearance of a candidate feature. Third, identify the first evidence that the maker treats the feature as consequential. Fourth, examine later interventions for counterfactual force: did they increase the probability that the feature, or a controlled transformation of it, would survive? Fifth, distinguish intervention from filtering by testing whether the maker can produce nearby alternatives while preserving the claimed invariant. Finally, compare the claimed human invariant against baseline tendencies of the system so that a model default is not mistaken for a user-originated relation.

This protocol changes the evidentiary object from an odometer to a sequence of commitments. It does not reward labor for its own sake. It can fail. A maker may claim to have controlled a relation and then be unable to preserve it under a neighboring edit. A history may show that the relation survived equally often without the maker’s intervention. Another system may reproduce the same supposedly personal invariant by default. These failures are useful because they make the account falsifiable.

The protocol also accommodates exploratory practices without demanding a master plan. An intention formed after an intermediate result is not disqualified merely because it is late. What matters is whether it becomes operationally consequential before the process ends. This is a narrower claim than saying that selection creates authorship, and a broader claim than saying that every authored element must have been specified before execution began.

9 Implications for authorship analysis

The immediate implication is methodological. Legal and cultural analysis should resist treating a technological class label as a production history. Two artifacts made with the same model can involve radically different distributions of initiative, constraint, delegated discretion, and later correction. Conversely, different tools can support the same structure of exploratory commitment.

For copyright, prospective stabilization does not resolve the statutory question of which expressive elements are protectable. The Copyright Office’s current position remains that present generally available systems usually leave too much expressive determination to the system when only textual direction is involved.[1] The proposed account instead identifies a fact pattern that future cases should be able to describe more accurately: not merely repeated generation, but later human interventions that cause a discovered expressive relation to persist across subsequent variation.

The proposal is also compatible with the negative lesson of *Midway*. A player who reacts skillfully to a game can have high causal leverage without originating the game’s recurring expressive structure.[6] Prospective stabilization would matter only when later human action establishes or preserves a relation that is not adequately explained by the system’s upstream expressive invariants alone. *Stern* therefore supplies an important companion test: identify what remains constant because of the system before crediting constancy to the user.[7]

The hardest cases will involve jointly conditioned features. A human may specify a relation coarsely, the model may elaborate it, and later human action may stabilize one of several elaborations. Element-level partition may then be more accurate than a single verdict about the whole artifact. *Andrien*’s distinction between rote embodiment and independent modification is useful here, but learned systems create a causal middle that older doctrine did not need to name.[8]

10 Discussion

The broader theoretical result is that control is not only a property of an initial command. It is a relation distributed over time among perception, available operations, system behavior, and retained state. A maker can acquire agency over a feature only after that feature becomes perceptible as a difference. The history of making therefore contains two kinds of branching: choosing among known

alternatives and creating the very distinctions by which later alternatives will be judged.

This temporal view also changes how accidents should be understood. Accident is often opposed to intention, but in exploratory making an accident can be the event that makes a new intention possible. The critical boundary is not accident versus control. It is whether the accident remains a terminal gift or becomes the object of subsequent accountable intervention.

The phrase “call the next shot” captures the difference. One need not predict every first occurrence. But after claiming that an unexpected relation matters, the maker should be able to expose that claim to the next transformation. Preserve the face while changing the light. Keep the spatial tension while reversing the gesture. Restore the relation after another operation damages it. If nothing the maker can do changes whether the relation survives, the claimed agency is weak. If later interventions reliably make the difference persist, the production history contains more than adoption.

11 Limitations

This argument has four important limits. First, it is not a statement of current U.S. law. The Copyright Office has not adopted prospective stabilization as an authorship test, and appellate courts have not resolved a richly documented case involving this fact pattern. Second, the account is evidentiary rather than normative. Showing causal creative agency does not establish that copyright should protect every resulting element.

Third, the proposal requires empirical validation. Different media may support very different forms of stabilization, and some forms of tacit practice may resist clean logging. The observer’s choice of what counts as the same feature across transformations can itself import contested aesthetic judgments. Fourth, co-adaptation complicates attribution. A user can become better at preserving a relation because the system has trained the user to prefer what the system already does well. The protocol therefore cannot replace comparative analysis of model defaults, alternative tools, and independent intention evidence.

12 Conclusion

The central mistake in many accounts of generative making is temporal. They ask whether a person specified the final expressive result before the system acted, or whether the person merely selected what the system supplied afterward. Exploratory practice occupies a third structure. The system produces an intermediate state; the maker discovers a difference; the difference becomes a commitment; later actions make that commitment consequential.

That structure does not erase machine contribution, stochasticity, or upstream model authorship. Nor does it turn every long iterative process into human authorship. It gives a smaller and more falsifiable question: after the unexpected feature appeared, could the maker make it matter to what happened next? A serious answer requires a path, not a class label, and a sequence of interventions, not an odometer. The first surprise may belong to the system, chance, the material, or no single agent. The next shot is where a new claim of agency begins.

References

- [1] U.S. Copyright Office. 2025. *Copyright and Artificial Intelligence, Part 2: Copyrightability*. January 2025. <https://www.copyright.gov/ai/Copyright-and-Artificial-Intelligence-Part-2-Copyrightability-Report.pdf>.
- [2] Pamela Samuelson, Christopher Jon Sprigman, and Matthew Sag. 2023. *Comments in Response to the Copyright Office’s Notice of Inquiry on Artificial Intelligence and Copyright*. October 30, 2023.
- [3] Donald A. Schon. 1992. “Designing as Reflective Conversation with the Materials of a Design Situation.” *Research in Engineering Design* 3: 131–147. doi:10.1007/BF01580516.
- [4] Shachar Don-Yehiya, Leshem Choshen, and Omri Abend. 2023. “Human Learning by Model Feedback: The Dynamics of Iterative Prompting with Midjourney.” *Proceedings of EMNLP 2023*: 4146–4161. doi:10.18653/v1/2023.emnlp-main.253.
- [5] Tobias Jung, Daniel Polani, and Peter Stone. 2011. “Empowerment for Continuous Agent–Environment Systems.” *Adaptive Behavior* 19(1): 16–39. doi:10.1177/1059712310392389.
- [6] *Midway Manufacturing Co. v. Artic International, Inc.*, 704 F.2d 1009 (7th Cir. 1983).
- [7] *Stern Electronics, Inc. v. Kaufman*, 669 F.2d 852 (2d Cir. 1982).
- [8] *Andrien v. Southern Ocean County Chamber of Commerce*, 927 F.2d 132 (3d Cir. 1991).

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